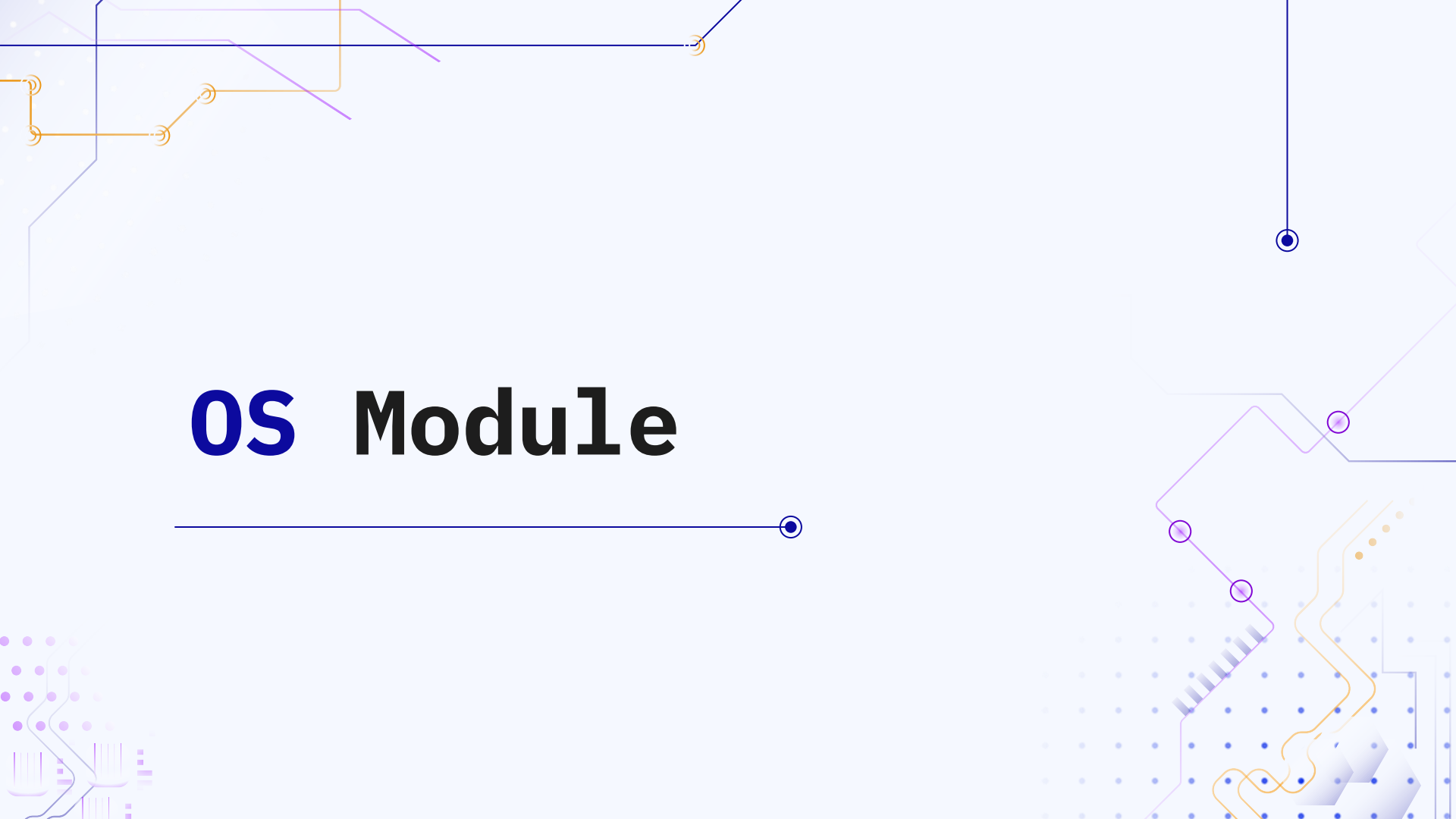


OS Module

The background features a light blue gradient with abstract circuit-like patterns. In the top-left, there are orange and purple lines with circular nodes. In the top-right, a blue line with a circular node is visible. The bottom-left has a cluster of purple dots and lines. The bottom-right contains a grid of blue dots with orange and purple lines winding through it, and a small cluster of orange dots. A horizontal blue line with a circular node is positioned below the title.

Introduction

The **os** module helps your Python program interact with the **operating system** (Windows, Linux, macOS). With it, you can:

- Work with **folders & files** (create, delete, rename, list)
- Handle **paths** (like C:\Users\... or /home/...)
- **Read/set** environment variables
- Get **current** working directory
- Do some **process-related tasks** (like getting process id)

Syntax

```
import os
```

Current Working Directory

os.getcwd() → Returns the folder where your Python script is currently operating.

Python

```
import os
print(os.getcwd())
```

os.chdir(path) → Move to a different folder

Python

```
import os
os.chdir(r"C:\Users\Amar\Documents") # Windows example (raw string)
print(os.getcwd())
```

⚠ If the path doesn't exist, Python throws an error.



Listing Files and Folders

Gives a list of names (files + folders) in a directory.

Python

```
import os
print(os.listdir(path="."))
```

You can loop through it:

Python

```
import os
for item in os.listdir(path=r"E:\Python Tutorials\Unit 14"):
    print(item)
```

Creating and Removing Directories

Create a folder

- `os.mkdir("folder")` → makes one folder
- `os.makedirs("a/b/c")` → makes nested folders (multi-level)

Python

```
import os
os.mkdir("notes")
os.makedirs("projects/python/os_module", exist_ok=True)
```

exist_ok → True prevents error if folder already exists.

Creating and Removing Directories

Remove a folder

- `os.rmdir("folder")` → removes an empty folder only
- For non-empty folders, use `shutil.rmtree()` (from `shutil` module)

Python

```
import os
os.rmdir("notes")
```

Python

```
import shutil
shutil.rmtree("projects") # deletes folder + everything inside
```

⚠ **rmtree()** deletes permanently. Use carefully.

File/Folder Existence Checks

Checks whether a file/folder exists.

- **`os.path.exists(path)`**

Python

```
import os
print(os.path.exists("notes"))      # True/False
print(os.path.exists("data.csv"))

print(os.path.isfile("data.csv"))   # True if it's a file
print(os.path.isdir("notes"))       # True if it's a folder
```

Path Handling

Paths differ across OS:

- **Windows:** C:\Users\Name\file.txt
- **Linux/macOS:** /home/name/file.txt

So instead of manually joining strings, use:

Syntax

```
os.path.join()
```

Python

```
import os
path = os.path.join("projects", "python", "app.py")
print(path)
```

Path Handling

Python

```
file_path = r"C:\Users\Amar\test\data.csv"

print(os.path.basename(file_path))    # data.csv
print(os.path.dirname(file_path))     # C:\Users\Amar\test
print(os.path.splitext(file_path))   # ('...\data', '.csv')
```

Best practice: Always use **os.path.join()** for portability.

Renaming and Deleting Files

Rename file/folder

Python

```
os.rename("old_name.txt", "new_name.txt")
```

Delete a file

Python

```
os.remove("temp.txt")
```

These operations are **permanent** (go to recycle bin or not depends on OS).
Use carefully.

File Metadata

`os.path.getsize(path)` → size in bytes

Python

```
size = os.path.getsize("data.csv")  
print(size, "bytes")
```

`os.path.getmtime()`, `getatime()`, `getctime()` & `os.stat()`

Python

```
p = r"E:\Python Tutorials\Unit 13\Projects\etesst.py"  
print(os.path.getmtime(p))  
print(os.path.getatime(p))  
print(os.path.getctime(p))  
print(os.stat(p))
```

Best Practices

- Prefer **os.path.join()** over manual path building
- Use **os.path.exists()** before delete/rename
- Use raw strings for Windows paths: **r"C:\path\file"**
- Handle errors using **try-except**

WATCH

Level up your coding with each episode in this focused Python series.



Next Video!

Pathlib
Module

